



LAUNCH COST ESTIMATE

Staging & Production: What It Costs to Actually Buy

A line-by-line, repo-grounded shopping list — infrastructure, SMS/email/monitoring accounts and payment-gateway setup — priced at current market rates, so you know what to purchase and roughly what to budget before going live.

Companion documents: deployment-plan.pdf (what to provision, Section 3) and launch-requirements.pdf (Section 5, business/service accounts) — this document adds pricing to both lists; it does not repeat their readiness checklists.

Pricing basis: researched market rates as of this document's preparation date, indicative FX $\approx$ ₹85 / US\$1. Treat every figure as an estimate to confirm on the vendor's site at time of purchase — see Section 6 for the full disclaimer.

1 How to Read This Estimate

Costs fall into three different shapes — mixing them into one number would be misleading, so they're kept separate throughout this document:

- **One-time / setup** — paid once (domain registration is actually annual, but only decided once): Section 3.
- **Fixed monthly infrastructure** — the same whether you have 10 students or 10,000, until you outgrow the tier: Section 4.
- **Usage-based** — scales with real traffic and revenue (SMS sent, emails sent, payment volume processed): Section 5.

Sizing assumption: the fixed-cost tiers below are sized for an early-launch cohort — comfortably up to a few thousand registered students and a handful of institutions. This is a starting budget, not a ceiling; Section 4's "Managed / HA" column is what you step up to once real traffic and uptime requirements justify it.

2 What Needs to Be Purchased

Nine things, in total — five infrastructure components already identified in deployment-plan.pdf §3, plus four external service accounts identified in launch-requirements.pdf §5:

#	Item	Type
1	Compute — a place to run the API, worker, student web and admin containers	Infra
2	Managed PostgreSQL	Infra
3	Managed Redis / Valkey	Infra
4	S3-compatible object storage (private, versioned)	Infra
5	Domain + DNS + TLS certificates	Domain / free TLS
6	SMS provider (MSG91) account + DLT entity registration	One-time + usage
7	Transactional email provider account	Usage-based
8	Error tracking (Sentry) account	Infra
9	Razorpay live-mode merchant account (KYC-approved)	Free setup + usage

3 One-Time & Setup Costs

Item	Estimated cost	Notes
Domain registration — rajyarank.in (1 year)	₹800 – ₹1,500 (~US\$10–18)	Registry wholesale is ~₹500/yr; budget the real renewal price, not a first-year teaser rate. Add ~₹1,000–1,500/yr for a matching .com if you want brand coverage.

Item	Estimated cost	Notes
DLT Principal Entity registration (required before any real SMS/OTP can send)	≈ ₹6,960 (~US\$82)	₹5,900 + 18% GST, one-time, per business entity, via the operator portal (Airtel SmartPing / Jio). Header and template registration are typically free.
SSL/TLS certificates	₹0	Let's Encrypt, auto-renewing — already wired into the deploy tooling per deployment-plan.pdf.
Razorpay account + live-mode KYC	₹0	No setup fee. Needs PAN/bank/business registration proof, and your Terms/Privacy/Refund pages live and linked (launch-requirements.pdf §2.3–2.4) before Razorpay approves live mode.
Secrets manager	₹0	A cloud-native secret store is bundled free with the hosting account at this scale.

Recommended professional services (procured by you, not fixed infra)

launch-requirements.pdf flags these as launch-blocking for a payments-and-PII platform. They're priced separately here because the range depends entirely on vendor and scope — get quotes before committing.

Item	Typical range	Notes
External penetration test (IDOR / role-escalation / upload abuse / webhook replay)	₹60,000 – ₹2,50,000 (~US\$700–2,950)	Boutique Indian security firms sit at the lower end; larger firms/CREST-certified testers cost more. Get 2–3 quotes.
Accessibility + performance audit	₹0 – ₹1,00,000 (~US\$0–1,175)	Automated tooling (Lighthouse CI, axe) is effectively free; an independent WCAG specialist audit costs more — your call on rigor before public launch.

4 Monthly Infrastructure — Fixed Costs

deployment-plan.pdf §3 deliberately left the container host as an open decision. Two honest paths, priced side by side — start Lean, move to Managed/HA once traffic and uptime needs justify it. Reference pricing: a mainstream managed-cloud provider (equivalent AWS/GCP pricing runs 20–40% higher for comparable specs).

Component	Staging	Production — Lean	Production — Managed / HA
Compute (API + worker + web + admin containers)	\$24 / ₹2,040	\$48 / ₹4,080	\$108 / ₹9,180
Load balancer / ingress	—	\$12 / ₹1,020	\$12 / ₹1,020
Managed PostgreSQL	\$15 / ₹1,275	\$60 / ₹5,100	\$120 / ₹10,200
Managed Redis / Valkey	\$15 / ₹1,275	\$15 / ₹1,275	\$30 / ₹2,550
Object storage (S3-compatible)	\$5 / ₹425	\$5 / ₹425	\$10 / ₹850
DNS + TLS	\$0	\$0	\$0
Error tracking (Sentry)	\$0 (free tier)	\$29 / ₹2,465 (Team)	\$80 / ₹6,800 (Business)

Component	Staging	Production — Lean	Production — Managed / HA
Transactional email — fixed floor	\$0 (free tier)	≈ \$5 / ₹425	≈ \$10 / ₹850
Monthly subtotal	≈ \$59 / ₹5,015	≈ \$174 / ₹14,790	≈ \$370 / ₹31,450

What each tier is for: *Staging* is always-on lean infrastructure for testing before every promotion to Production (per deployment-plan.pdf's promotion flow). *Production — Lean* is a single right-sized VM plus managed data services — a sensible real-launch starting point. *Managed / HA* swaps the VM for a multi-node cluster with standby database/cache — reach for this once you have real paying users depending on uptime.

5 Usage-Based Costs — Scale with Traffic & Revenue

These are not purchased upfront — they're billed by consumption, so they're shown here as a rate plus a worked example, not a fixed line item.

Service	Rate	Worked example
Razorpay (domestic cards / UPI)	2% + 18% GST (≈2.36% effective)	₹10,00,000/month in course sales → ≈ ₹23,600/month in gateway fees. No setup or monthly fee — pure percentage of revenue collected.
SMS OTP (MSG91)	₹0.15 – ₹0.20 per OTP	10,000 logins/signups/month → ₹1,500–2,000 (~US\$18–24)/month. Institution-enrolled students set up directly by their Academic Head skip phone OTP entirely — a real cost lever, not just a UX choice.
Transactional email (AWS SES, beyond the free tier)	\$0.10 per 1,000 emails	50,000 emails/month ≈ \$5 (₹425)/month — negligible until volume is very high.

6 What to Actually Set Aside

Combining Staging + Production (Lean tier) — the realistic starting point for a first real launch — before usage-based SMS/email/Razorpay fees, which depend on actual traffic and revenue.

One-time setup (domain + DLT entity registration)	≈ ₹8,460 (~\$100)
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First month, Staging + Production-Lean recurring	≈ ₹19,805 (~\$233)
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Total to go live (Month 1)	≈ ₹28,265 (~\$333)
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Steady-state monthly, Staging + Production-Lean	≈ ₹19,805 (~\$233)
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Annualised (fixed infra only)	≈ ₹2,37,660 (~\$2,795)
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Not included in the totals above: the recommended professional services from Section 3 (₹60,000–₹3,50,000, one-time, whenever you're ready for public launch — not required to stand up Staging) and the usage-based costs from Section 5, which only start accruing once real students are actually using the platform.

7 Assumptions & Disclaimer

- Prices reflect one representative mainstream cloud provider and one representative India SMS/payments vendor, researched at the time this document was prepared. Actual vendor pricing changes — reconfirm on the vendor's own pricing page immediately before purchasing.
- The ₹85/US\$1 exchange rate used throughout is approximate and will drift; USD-billed services (compute, database, Sentry, SES) will move with it.
- Fixed-tier sizing assumes an early-launch cohort, not a specific guaranteed student count — real usage should be monitored against these tiers and upgraded proactively, not reactively.
- Razorpay's 2% domestic rate is its standard published rate; high-volume merchants (₹5–10 lakh+/month) can typically negotiate custom Enterprise pricing directly with Razorpay.
- This document estimates *infrastructure and third-party service* costs only — it does not include engineering time, content production, marketing spend, or the professional-services items in Section 3 beyond the ranges already stated.

Ready to purchase

Section 2's nine items are the actual shopping list. Sections 3–5 are what to budget for each. Once accounts exist and credentials are in hand, [deployment-plan.pdf](#) §5–8 covers wiring them in and deploying.

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